FAHAD WAHAB

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Early-career Machine Learning engineer building end-to-end LLM and computer vision systems. Hands-on experience in model training, evaluation, semantic retrieval, and real-time deployment on edge devices. Strong foundation in production-oriented ML pipelines, modular system architecture, and API-driven integration.

SKILLS

- **Machine Learning & LLM Engineering**: Machine Learning Pipeline, Natural Language Understanding (NLU), Large Language Models (LLMs), Prompt Engineering, Retrieval Augmented Generation (RAG), Model Training, Model Evaluation, ML Pipelines
- **Programming & Tools**: Python, TensorFlow, Keras, PyTorch, NumPy, OpenCV, scikit-learn, ROS, Git (Version Control), Docker, n8n (Workflow automation), CI/ CD Fundamentals
- **Data & Systems**: Data Processing, Vector Databases (FAISS, Chroma), Modular System Architecture
- **Infrastructure & Deployment**: Docker, SSH Deployment, REST API Integration, Edge & Real time Inference, Cloud Fundamentals (AWS Basics)

EDUCATION

MASTERS IN E-MOBILITÄT (ARTIFICIAL INTELLIGENCE) — **ARPIL 2025 TO MARCH 2026**

Friedrich-Alexander-Universität – Erlangen-Nürnberg, Germany

BACHELOR OF MECHANICAL ENGINEERING — **SEPTEMBER 2014 TO JUNE 2018**

National University of Science and Technology – Islamabad, Pakistan

PROJECTS

MASTER'S THESIS — **OCTOBER 2025 TO MARCH 2026**

EDGE-DEPLOYED CONVERSATIONAL AI WITH RETRIEVAL-AUGMENTED GENERATION (RAG)

- Designed and implemented a modular 5-stage conversational AI pipeline integrating STT, LLM-based NLU, RAG, and TTS.
- Applied prompt engineering and FAISS-based semantic retrieval (k=4) with sub-100ms latency for grounded response generation.
- Architected API-driven components and containerized the system using Docker with CI/CD-ready deployment workflows.
- Deployed and optimized the solution on NVIDIA Jetson for low-latency real-time edge inference.

MASTER'S PROJECT — **JANUARY 2025 TO AUGUST 2025**

CNN-BASED REAL-TIME INFANT SLEEP STATE CLASSIFICATION

- Designed and implemented a real-time computer vision pipeline integrating MTCNN/dlib based facial landmark detection with MobileNet-based CNN classification for infant sleep-state inference.
- Trained a MobileNet-based CNN using transfer learning with regularization and dropout to improve generalization and reduce overfitting.
- Processed video at 15 FPS and implemented a 90-second temporal confirmation window to ensure stable real-time predictions.

PROFESSIONAL EXPERIENCE

MAINTENANCE ENGINEER — **NOVEMBER 2018 TO MAY 2022**

CORPS OF ELECTRICAL AND MECHANICAL ENGINEERS, PAKISTAN

- Led 50 person team, supervised system diagnostics, reliability assessments, and technical reporting.
- Analyzed ECU and vehicle system data to detect anomalies and support decision-making.
- Delivered training on hybrid and modern automotive systems, improving team capability.

CERTIFICATIONS

- Getting Started with AI on Nvidia Jetson Nano **(NVIDIA)**
- Generative AI with Large Language Models **(AWS, DeepLearning.AI, Coursera)**
- Deep Learning Specialization **(DeepLearning.AI, Coursera)**